Types of Vegetables in Cambodia

Horn Sarun[1] *

[1]Royal University of Agriculture, Phnom Penh, Cambodia.

Received: 1 November 2025/Revised: 1 December 2025/ Accepted: 1 January 2026/ Published online: 9 June 2026

* Corresponding: hornsarun@gmail.com

Abstract

Vegetables are a vital component of Cambodian food systems, contributing to household nutrition, rural livelihoods, and smallholder income generation. Grown mainly in home gardens, peri-urban farms, and small-scale commercial systems, vegetables in Cambodia are highly diverse and adapted to tropical lowland environments and seasonal rainfall. This review classifies vegetables commonly produced in Cambodia based on edible plant parts, outlines their nutritional and economic importance, and discusses key challenges and opportunities for sustainable vegetable production. Strengthening vegetable systems is essential for improving food security, nutrition, and resilience of smallholder farming systems in Cambodia.

Keywords: Vegetables; Cambodia; Leafy vegetables; Root and tuber crops; Food security; Smallholder farming

How to Cite: Sarun, H. (2026). Types of Vegetables in Cambodia. *Journal of Agriculture and Environment*, *3*(3), 535-539.

1. Introduction

Vegetables play an essential role in Cambodian diets, rural livelihoods, and smallholder farming systems. They contribute significantly to household nutrition, income generation, and food security, particularly through home gardens and small-scale commercial production (Sarun, 2026a; Keats et al., 2018). Vegetable production in Cambodia is largely characterized by diverse species adapted to tropical lowland conditions and seasonal rainfall patterns (Sarun, 2026b).

2. Classification of Vegetables in Cambodia

Vegetables in Cambodia can be broadly classified based on the **edible plant part**, a system widely used in horticultural science (Hartmann & Kester, 1983, Sarun, 2026c).

2.1 Leafy vegetables

Leafy vegetables are the most commonly consumed and widely grown vegetables in Cambodia (Sarun, 2026c). Typical examples include water spinach (*Ipomoea aquatica*), mustard greens (*Brassica juncea*), Chinese kale (*Brassica oleracea* var. *alboglabra*), and amaranth (*Amaranthus* spp.). These crops are fast-growing and well suited to home gardens and peri-urban farming systems Ty et al., 2013; Sarun, 2026d).

2.2 Fruit vegetables

Fruit vegetables are plants in which the botanical fruit is consumed. Common fruit vegetables in Cambodia include eggplant (*Solanum melongena*), tomato (*Solanum lycopersicum*), chili (*Capsicum* spp.), cucumber (*Cucumis sativus*), pumpkin (*Cucurbita* spp.), and bitter gourd (*Momordica charantia*) (Sarun, 2026e). These crops are important cash crops for smallholders supplying local markets (Sreynech et al., 2021; Sarun, 2026f).

2.3 Root and tuber vegetables

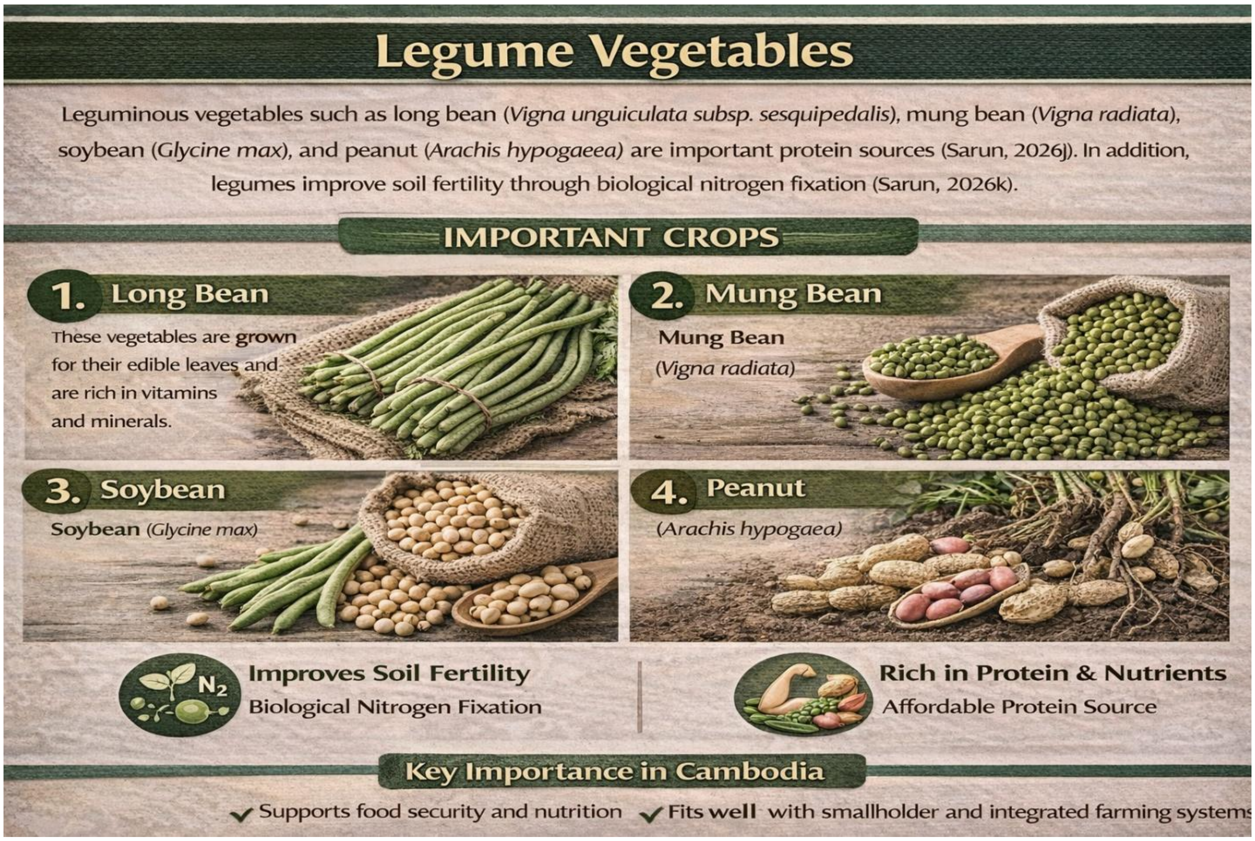
Root and tuber vegetables such as cassava (*Manihot esculenta*), sweet potato (*Ipomoea batatas*), taro (*Colocasia esculenta*), carrot (*Daucus carota*), and radish (*Raphanus sativus*) are widely cultivated (Sarun, 2026g). They are valued for their high energy content and tolerance to drought and poor soils, making them important for food security (Masalkar & Keskar, 1998; Sarun, 2026h).

2.4 Bulb and stem vegetables

Bulb and stem vegetables include onion (*Allium cepa*), garlic (*Allium sativum*), shallot (*Allium ascalonicum*), and celery (*Apium graveolens*) (Sarun, 2026h). These crops are widely used in Cambodian cuisine as flavoring agents and are often grown in irrigated dry-season systems (Sarun, 2026i).

2.5 Legume vegetables

Leguminous vegetables such as long bean (*Vigna unguiculata* subsp. *sesquipedalis*), mung bean (*Vigna radiata*), soybean (*Glycine max*), and peanut (*Arachis hypogaea*) are important protein sources (Sarun, 2026j). In addition, legumes improve soil fertility through biological nitrogen fixation (Sarun, 2026k).



2.6 Indigenous and wild vegetables

Cambodia also utilizes a wide range of indigenous and semi-wild vegetables, including banana flower (*Musa* spp.), sesbania flower (*Sesbania grandiflora*), water mimosa (*Neptunia oleracea*), and various wild leafy greens (Sarun, 2026l). These vegetables are culturally important and often more resilient to climate variability (Keats et al., 2018; Sarun, 2026m).

3. Importance of Vegetables in Cambodia

Vegetables contribute to **nutrition**, **income diversification**, and **livelihood resilience**. Home garden vegetables improve dietary diversity and micronutrient intake, particularly for women and children (Jindal & Dhaliwal, 2017; Sarun, 2026n). Commercial vegetable production offers higher returns per unit area than staple crops and supports rural employment (Sarun, 2026o).

4. Challenges and Opportunities

Vegetable production in Cambodia faces constraints including seasonality, pest and disease pressure, limited irrigation, and food safety concerns. However, opportunities exist through improved extension services, protected cultivation, organic and safe-vegetable certification, and integration into agroforestry and home garden systems (Keats et al., 2018).

5. Conclusion

Cambodia grows a diverse range of vegetables that can be classified into leafy, fruit, root and tuber, bulb, legume, and indigenous types. Strengthening vegetable production systems through improved management and market integration can enhance nutrition, farmer incomes, and national food security.

References

Hartmann, H. T., & Kester, D. E. (1983). *Plant propagation: principles and practices* (pp. 727-pp).

Jindal, S. K., & Dhaliwal, M. S. (2017). Development of vegetable nutrition garden model for diet diversification and improved nutrition security of urban and peri-urban households. *International Journal of Horticulture*, *7*.

Keats, S., Sim, S., & Phon, D. (2018). Country report: scaling up home gardens for food & nutrition security in Cambodia.

Masalkar, S. D., & Keskar, B. G. (1998). Other roots, tubers, and rhizomes. In *Handbook of Vegetable Science and Technology* (pp. 159-188). CRC Press.

Sarun, H. (2026a). Principles and Practices of Plant Breeding — A Comprehensive Review. *Journal of Agriculture and Environment*, *3*(2), 460-465.

Sarun, H. (2026b). Principles of Plant Breeding — A Comprehensive Review. *Journal of Agriculture and Environment*, *3*(3), 466-470.

Sarun, H. (2026c). Plant Propagation: Principles & Practices — A Nano Review. *Journal of Agriculture and Environment*, *3*(3), 471-475

Sarun, H. (2026d). Principles and Classification of Plant Propagation Methods. *Journal of Agriculture and Environment*, *3*(3), 476-481.

Sarun, H. (2026e). Seed Propagation: Physiology, Dormancy, and Germination Mechanisms. *Journal of Agriculture and Environment*, *3*(3), 482-487.

Sarun, H. (2026f). Vegetative Propagation Using Cuttings: Types, Techniques, and Applications. *Journal of Agriculture and Environment*, *3*(3), 488-492.

Sarun, H. (2026g). Layering Methods in Plant Propagation and Their Practical Importance. *Journal of Agriculture and Environment*, *3*(3), 493-497.

Sarun, H. (2026h). Grafting and Budding: Compatibility, Techniques, and Horticultural Uses. *Journal of Agriculture and Environment*, *3*(3), 498-500.

Sarun, H. (2026i). Micropropagation and Tissue Culture in Modern Plant Production. *Journal of Agriculture and Environment*, *3*(3), 501-504.

Sarun, H. (2026j). Clonal Propagation for Maintaining True-to-Type Crop Varieties. *Journal of Agriculture and Environment*, *3*(3), 505-509.

Sarun, H. (2026k). Role of Plant Growth Regulators in Vegetative Propagation. *Journal of Agriculture and Environment*, *3*(3), 510-514.

Sarun, H. (2026l). Plant Propagation Techniques for Forestry and Agroforestry Systems. *Journal of Agriculture and Environment*, *3*(3), 515-519.

Sarun, H. (2026m). Plant Propagation in Conservation of Rare and Endangered Plant Species. *Journal of Agriculture and Environment*, *3*(3), 520-524.

Sarun, H. (2026n). Plant Propagation and Nursery Management. *Journal of Agriculture and Environment*, *3*(3), 525-529.

Sarun, H. (2026o). Types of Plant Propagation and Their Importance. *Journal of Agriculture and Environment*, *3*(3), 530-534.

Sreynech, O. U. C. H., ROS, D., SAKHAN, S., THUN, V., OUK, M., TANAKA, K., & KAWAZU, Y. Exploration and Collection of Plant Genetic Resources in Northeastern Cambodia, 2021.

Ty, C., Sina, V., Borin, K., & Preston, T. R. (2013). Effect of different levels of biochar on the yield and nutritive value of Celery cabbage (Brassica chinensis var), Chinese cabbage (Brassica pekinensis), Mustard green (Brassica juncea) and Water spinach (Ipomoea aquatica). *Livest Res Rural Dev*, *25*.